

Knowledge Graph

Joemon M Jose

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Motto Latin: Via, Veritas, Vita
Motto in English The Way, The Truth, The Life
Type Public research university · Ancient university
Established 7 January 1451
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The University of Glasgow is a public research university in Glasgow, Scotland. Founded by papal bull in 1451, it is the fourth-oldest

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Public university in Glasgow, Scotland

The University of Glasgow is a public research university in Glasgow, Scotland. Founded by papal bull in 1451 (O.S. 1450), it is the fourth-oldest university in the English-speaking world and one of Scotland's four ancient universities. Wikipedia

Address: Glasgow G12 8QQ
Total enrollment: 26,635 (2017)
Founded: 1451
Rector: Rita Rae, Lady Rae
Undergraduate tuition and fees: Local tuition 1,820 GBP; Domestic tuition 9,000 GBP; International tuition 16,350 GBP (2015 - 16)
Chancellor: Dame Katherine Granger
Phone: 0141 330 2000
Colours: More: Arts Dentistry Divinity Engineering Law Medicine Nursing Science Social Sciences Veterinary Medicine
Hours: Open 24 hours
Endowment: £234.3 million (2023)
Motto: Latin: Via, Veritas, Vita
Affiliations: ACU; EUA; The Guild; PEGASUS; Russell Group; TPC; UANet; Universitas 21; Universitas Scotland; Universities UK
Budget: £944.2 million (2022/23)
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Tue, 5 Mar 09:00 PG Open Day

Tue, 5 Mar 18:10 Choral Evensong

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The University of Glasgow is a major research-led university

Barack Obama

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barack obama

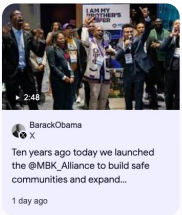
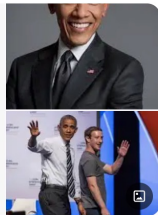
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Barack Obama

44th U.S. President



Age
62 years

Siblings
Malik Obama, Maya Soetoro-Ng, George Obama, Auma...

BarackObama
X
Ten years ago today we launched the @MBK_Alliance to build safe communities and expand...
1 day ago

The White House ...
Barack Obama | The White House

Wikipedia

https://en.wikipedia.org/wiki/Barack_Obama

Barack Obama

Barack Hussein Obama II is an American politician who served as the 44th president of the United States from 2009 to 2017. A...
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The White House (.gov)
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Barack Obama

Barack Obama served as the 44th President of the United States. His story is the American story — values from the heartland, a middle-class upbringing in a ...

About

barackobama.com

Barack Hussein Obama II is an American politician who served as the 44th president of the United States from 2009 to 2017. A member of the Democratic Party, he was the first African-American president in U.S. history. [Wikipedia](#)

Born: 4 August 1961 (age 62 years), Kapiʻolani Medical Center for Women & Children, Honolulu, Hawaii, United States

Siblings: Malik Obama, Maya Soetoro-Ng, George Obama, MORE

Presidential term: 20 January 2009 – 20 January 2017

Party: Democratic Party

Vice president: Joe Biden (2009–2017)

Children: Malia Ann Obama, Sasha Obama

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Barack Obama

44th President of the USA

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Quick Facts
Born: Barack Hussein Obama II · August...
Political party: Democratic
Spouse: Michelle Robinson (m. 1992)
Children: Malia · Sasha
Parents: Barack Obama Sr. · Ann Dunham

Early life and career
Obama was born on August 4, 1961, at Kapiolani Medical Center for Women and Children in Honolulu, Hawaii. He is the only president born outside the contiguous 48 states. He was born to an American mother and a Kenyan... See more

Legal career
He joined Davis, Miner, Barnhill & Galland, a 13-attorney law firm specializing in civil rights litigation and neighborhood economic development, where he was an *associate* for three years from 1993 to 1996, then of *counsel* from ... See more

Barack Obama - Wikipedia

https://en.wikipedia.org/wiki/Barack_Obama

Barack Hussein Obama II is an American politician who served as the 44th president of the United States from 2009 to 2017. A member of the Democratic Party, he was the first African-American president in U.S. history. Obama previously served as a U.S. senator representing Illinois from 2005 to 2008, as an Illinois state senator from 1997 to 20... See more

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News about Barack Obama

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Former President of the United States



Kamala Harris

Vice President Of The United States

Current Position
Former President of the United States Vice President Of The United States

Education Level
Bachelor's and law degrees Bachelor's and law degrees

Previous Position
Senator from Illinois Senator from California

If you're walking down the right path and you're willing to keep walking, eventually you'll make progress.

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Knowledge Graph!



Knowledge graph

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Knowledge graphs (KGs) organise data from multiple sources, capture information about entities of interest in a given domain or task (like people, places or events), and forge connections between them.

T

The Alan Turing Institute

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Knowledge graphs | The Alan Turing Institute

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Wikipedia

https://en.wikipedia.org/wiki/Knowledge_graph/

Knowledge graph

A digital structure that represents knowledge as concepts and the relationships between them (facts). A knowledge graph can include an ontology that flows both ...

History Definitions Using a knowledge graph for ... Entity alignment



Onto2net

https://www.onto2net.com/knowledge-graph/fundamentals/

What is a Knowledge Graph?

Knowledge graphs are a collection of interlinked descriptions of entities that put data into context and enable data integration, analytics & sharing.



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▼

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▼

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IBM

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What is a Knowledge Graph?

A knowledge graph represents a network of real-world entities—such as objects, events, situations or concepts—and illustrates the relationship between them.

What is a knowledge graph? How a knowledge graph works

Google Blog

https://blog.google/products/search/reintroducing-k/

Introducing the Knowledge Graph: things, not strings

16 May 2012 — The Knowledge Graph enables you to search for things, people or places that Google knows about—landmarks, celebrities, cities, sports teams, ...

Google Help

https://support.google.com/knowledgegraph/answer/

How Google's Knowledge Graph works

Google's search results sometimes show information that comes from our Knowledge Graph, our database of billions of facts about people, places, and things.

Stan4og

https://www.stan4og.com/knowledge-graph/

What is a Knowledge Graph

A flexible data layer. A Knowledge Graph, with its ability to make real-world content machine-understandable, is the ideal tool for enterprise data integration.



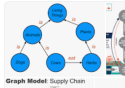
AlterSoft

https://www.altersoft.com/blog/knowledge-graph/

An Introduction to Knowledge Graphs

1 Oct 2022 — A knowledge graph is a new to integrate data coming from a variety of





Graph Model: Supply Chain



Knowledge graph

In knowledge representation and reasoning, a knowledge base is a knowledge base that uses a graph-structured data model or topology to represent and operate on data. Wikipedia

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Knowledge graph

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Knowledge graph - Wikipedia

https://en.wikipedia.org/wiki/Knowledge_graph

In knowledge representation and reasoning, a **knowledge base** that uses a **graph-structured data model or topology to represent and operate on data**. Knowledge graphs are often used to store interlinked descriptions of entities – objects, events, situations or abstract concepts — . See more

History

The term was coined as early as 1972 by the Austrian logician Edgar W. Schnöder, in a discussion of how to build modular institutional systems for courses. In the late 1980s, the University of Groningen and University of Twente jointly . See more

Using a knowledge graph for reasons

A knowledge graph formally represents semantics by describing entities and their relationships. Knowledge . See more

See also

- Concept map
- Diagram showing

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Knowledge graph

Knowledge base

In knowledge representation and reasoning, a knowledge graph is a knowledge base that uses a graph-structured data model or topology to represent and operate on data. Knowledge graphs are often used to ...

W

Knowledge graphs available on the Internet form a **Linked Open Data Cloud** (LOD Cloud), semantically combining multiple graphs into one giant network.



One of the most powerful resources for a knowledge graph search is **natural language processing (NLP)**. NLP enables search to recognize entities in the graph as well as user intent.

Google's knowledge Graph has millions of connected data pieces describing real-world entities like people, places, and things and forming the graph's nodes.

Explore more







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Images of Knowledge Graph

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What is a Knowledge Graph? | Ontatlas...

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What's the Difference Between an Ontology and a Knowledge Graph?

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IBM

<https://www.ibm.com/topics/knowledge-graph>

What is a Knowledge Graph? | IBM

Was a knowledge graph is a network of real-world entities and their relationships, stored in a graph database and visualized as a graph structure. Learn how knowledge graphs work.

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K PDF Knowledge Graphs - ResearchGate

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Neelg

<https://neelg.com/blog/what-is-knowledge-graph/>

What is a Knowledge Graph? - Graph Database & Analytics



was: 18 Jul 2023 A knowledge graph is a means to capture and explore information about foundational (business) entities in a domain or a business and the relationships between ...

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I An Introduction to Knowledge Graphs | SAS Blog

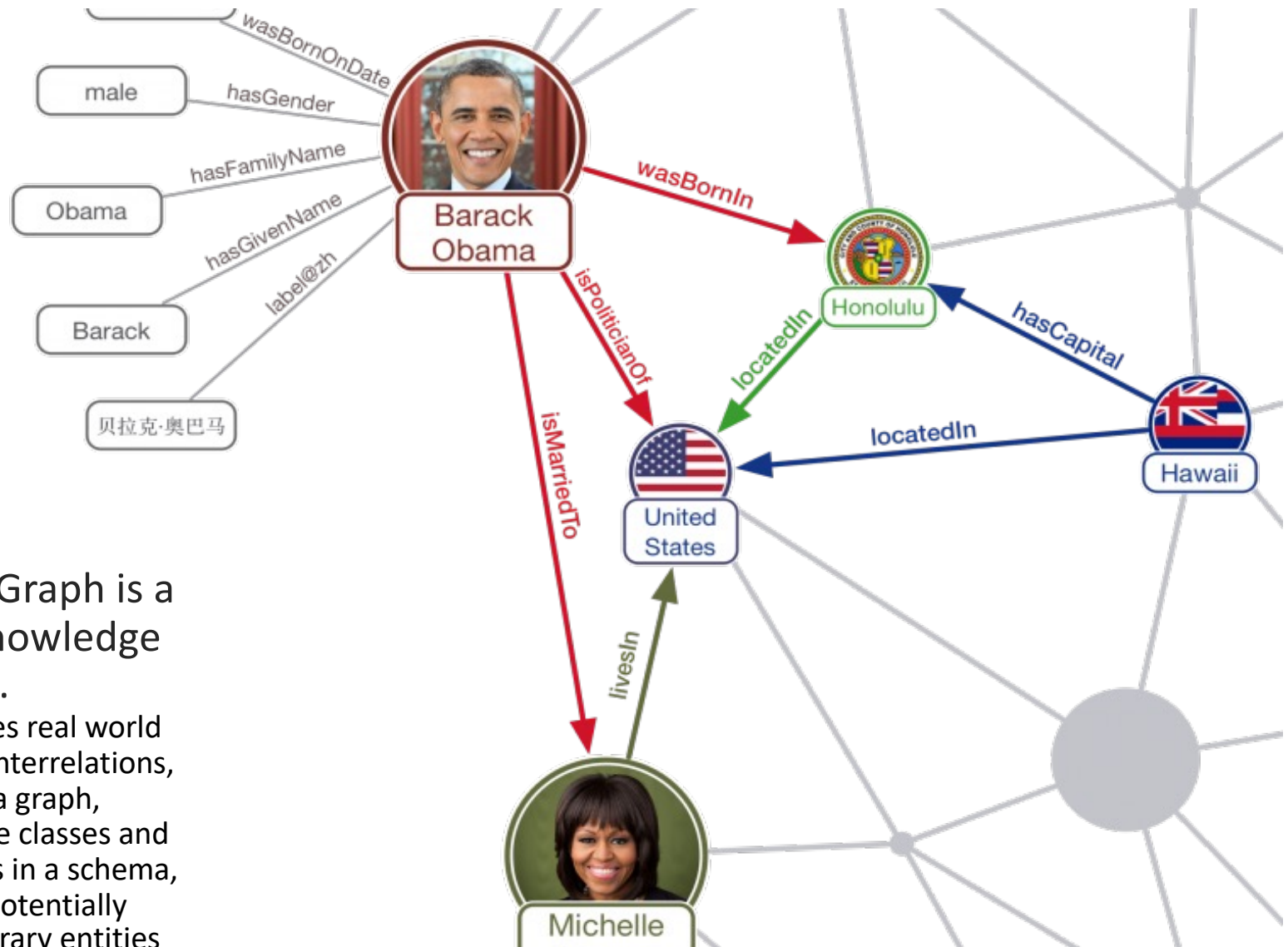
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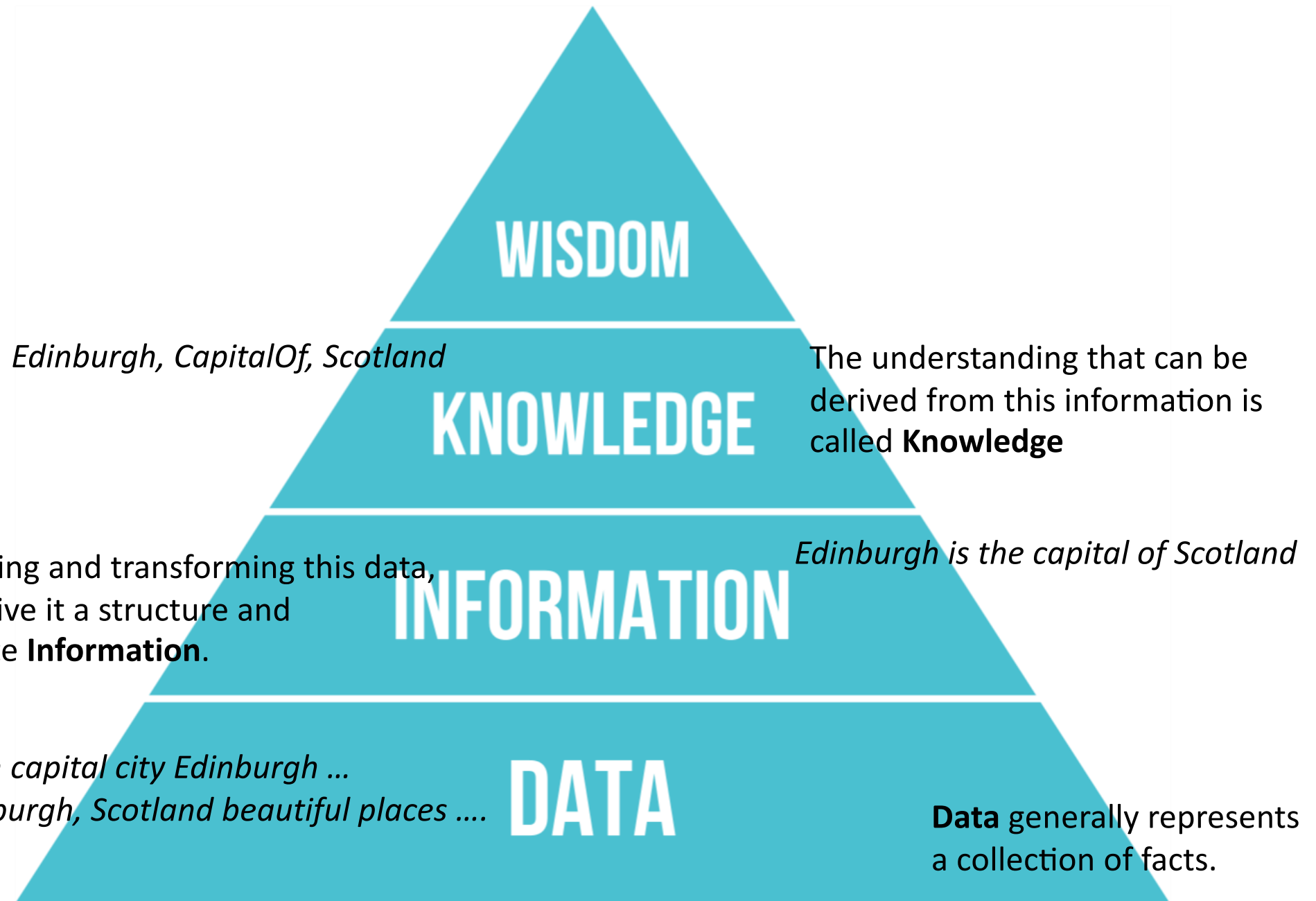
Knowledge graphs

- Knowledge graphs are one of the most fascinating concepts in data science
- Knowledge graphs are large networks of entities, their semantic types, properties, and relationships between entities.
 - Kroetsch & Weikum, 2016
 - Journal of Web Semantics Special Issue on Knowledge Graphs



A Knowledge Graph is a structured Knowledge Base.

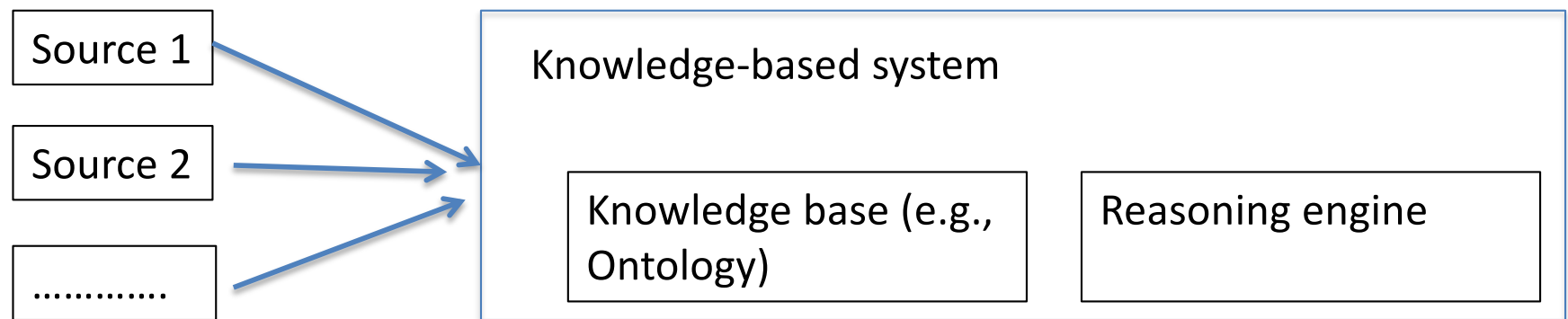
- (i) mainly describes real world entities and their interrelations, organized in a graph,
- (ii) defines possible classes and relations of entities in a schema,
- (iii) allows for potentially interrelating arbitrary entities with each other and
- (iv) covers various topical domains



Refined definition – Knowledge Graph

- A knowledge graph acquires and integrates information into an ontology and applies a reasoner to derive new knowledge
- Architecture

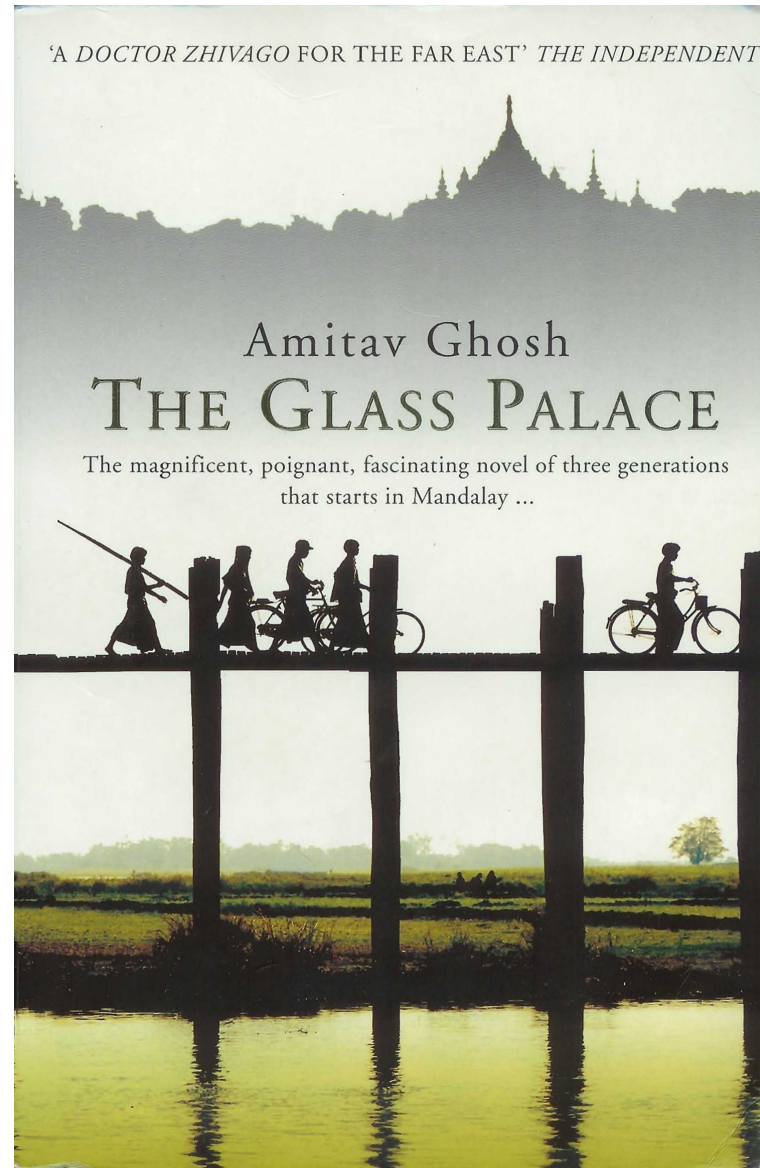
What is a knowledge graph?



knowledge graph vs ontology

- A knowledge graph is a **representation** of interconnected concepts, often derived from data sources.
- Ontologies are often **carefully crafted**, with human experts specifying the relationships and constraints.
 - Unique vocabulary
 - Set of relationships
- Knowledge graphs **rely on ontologies** to organize and structure data.

Let us look at a book...



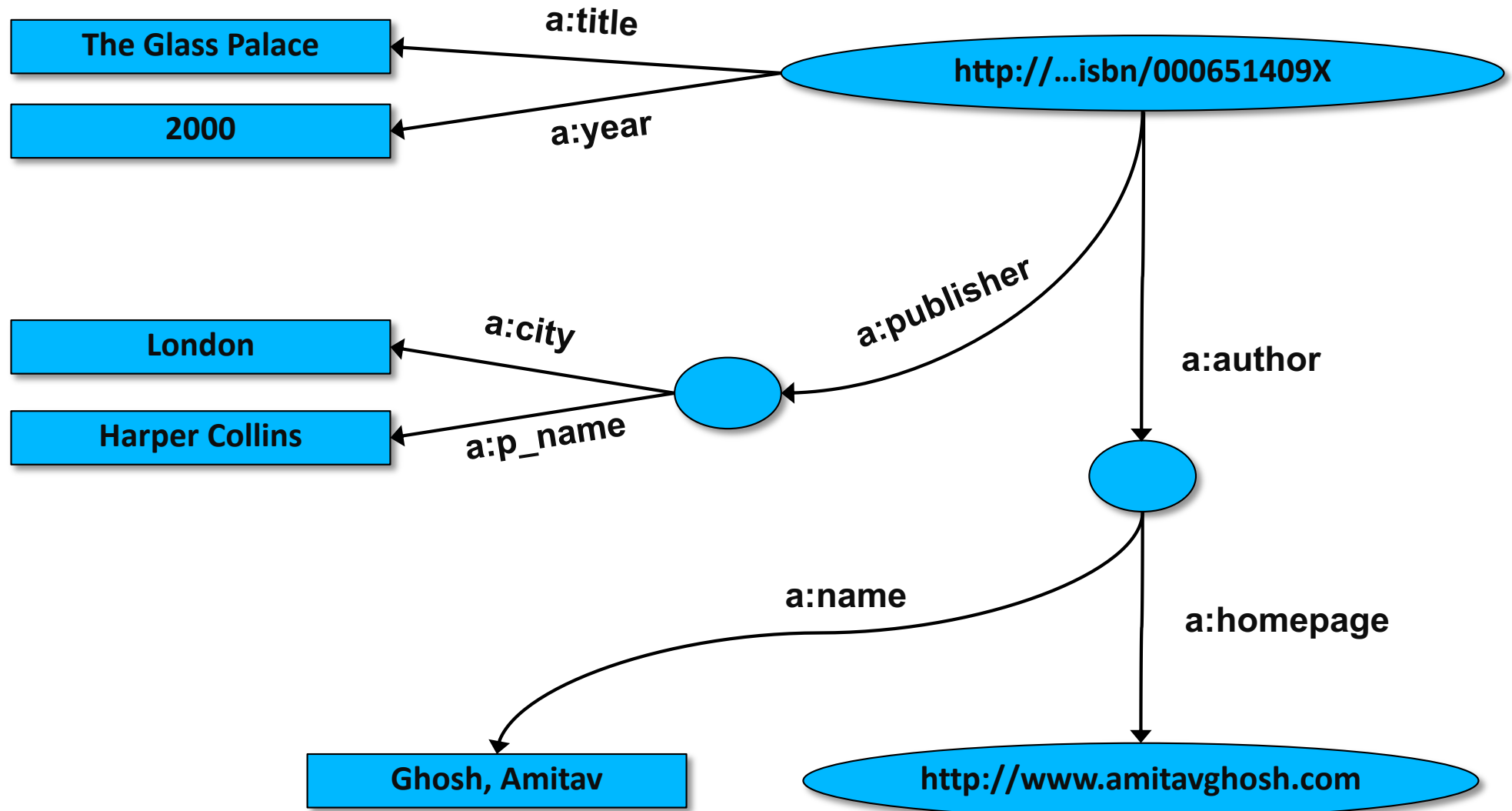
A simplified bookstore data

ID	Author	Title	Publisher	Year
ISBN 0-00-6511409-X	id_xyz	The Glass Palace	id_qpr	2000

ID	Name	Homepage
id_xyz	Ghosh,Amitav	http://www.amitavghosh.com

ID	Publisher's name	City
id_qpr	Harper Collins	London

export your data as a set of relations

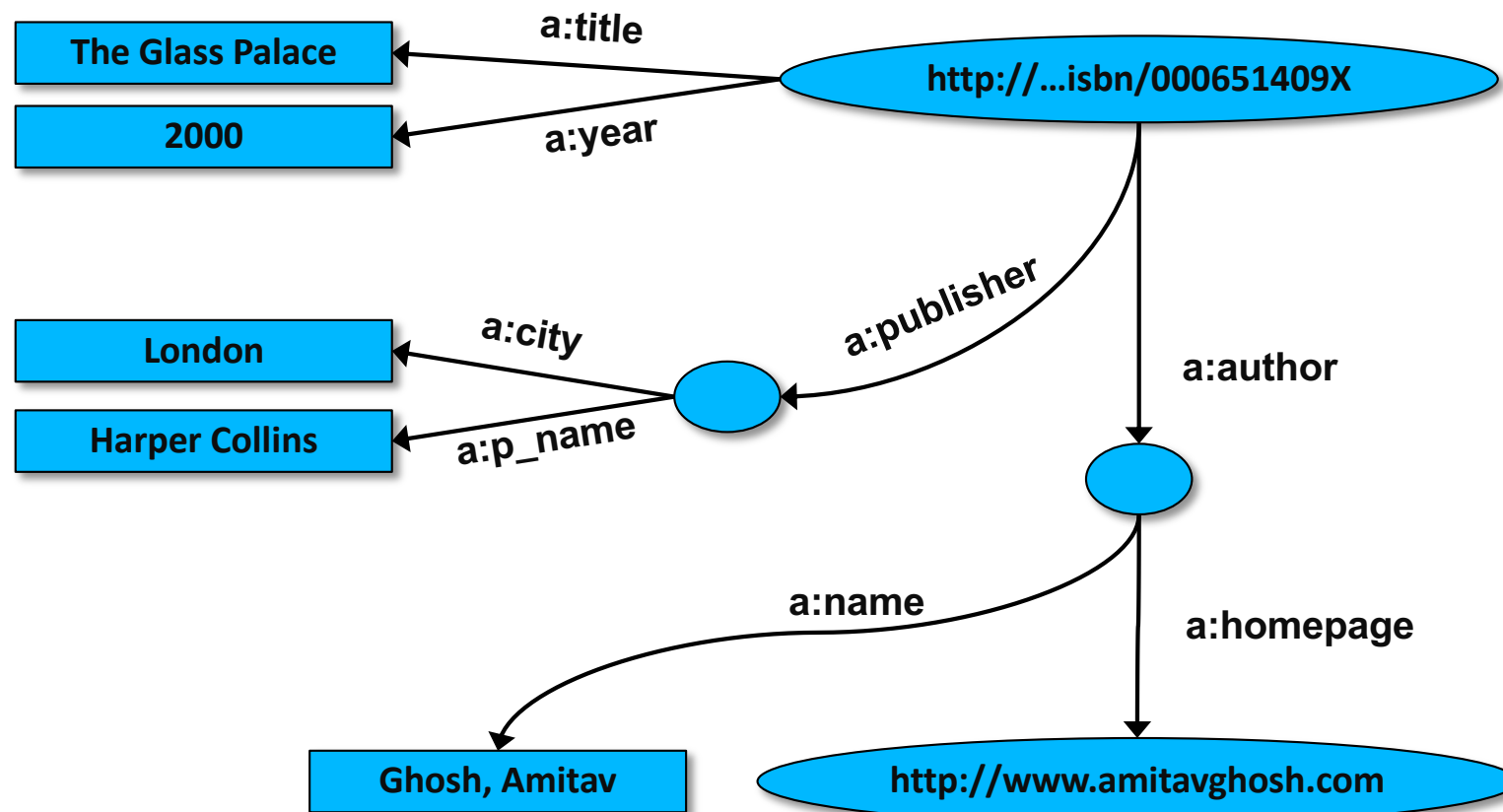


Bookstore

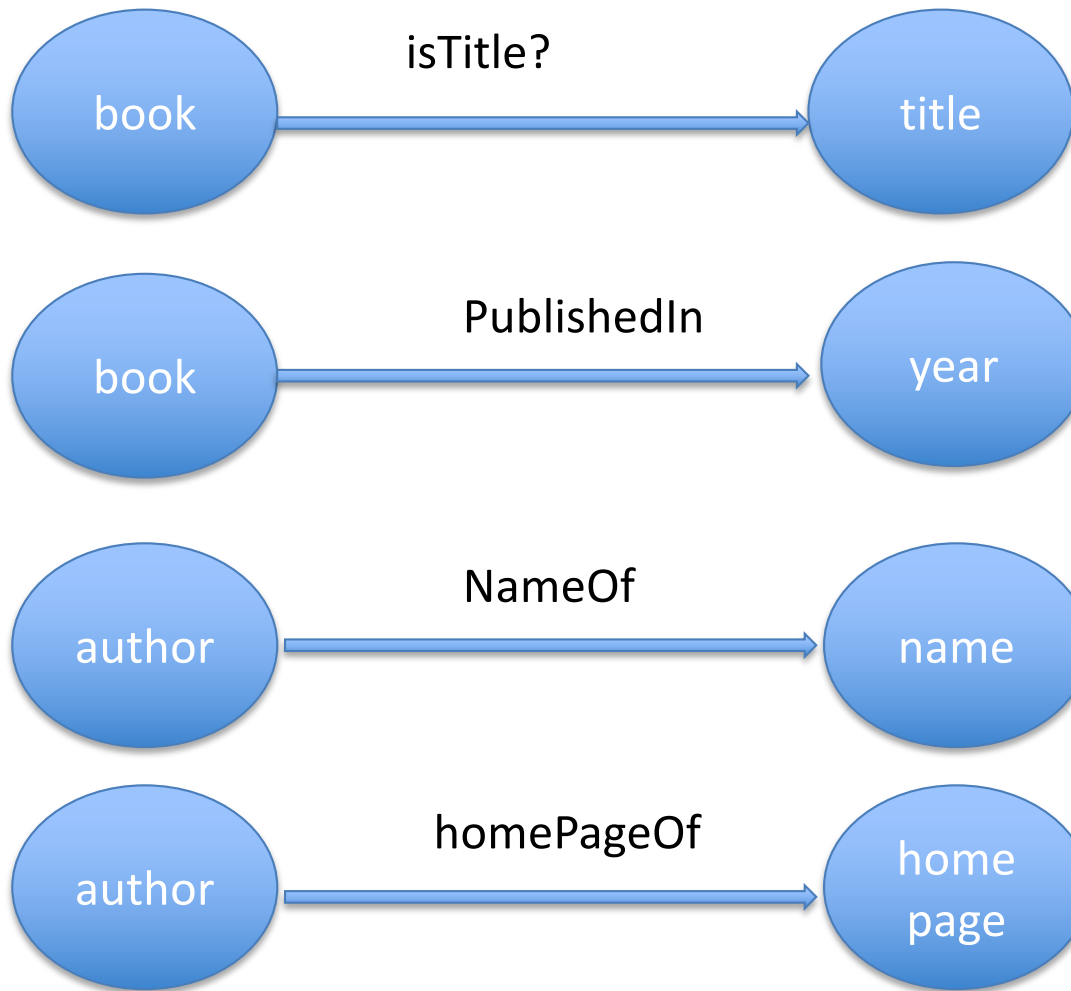
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ID	Name	Homepage
id_xyz	Ghosh, Amitav	http://www.amitavghosh.com

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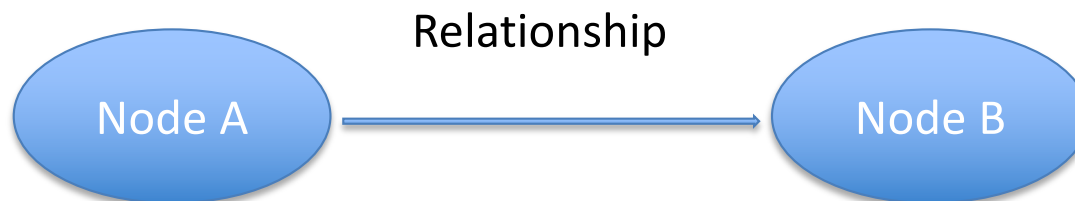


Subject, predicate, object



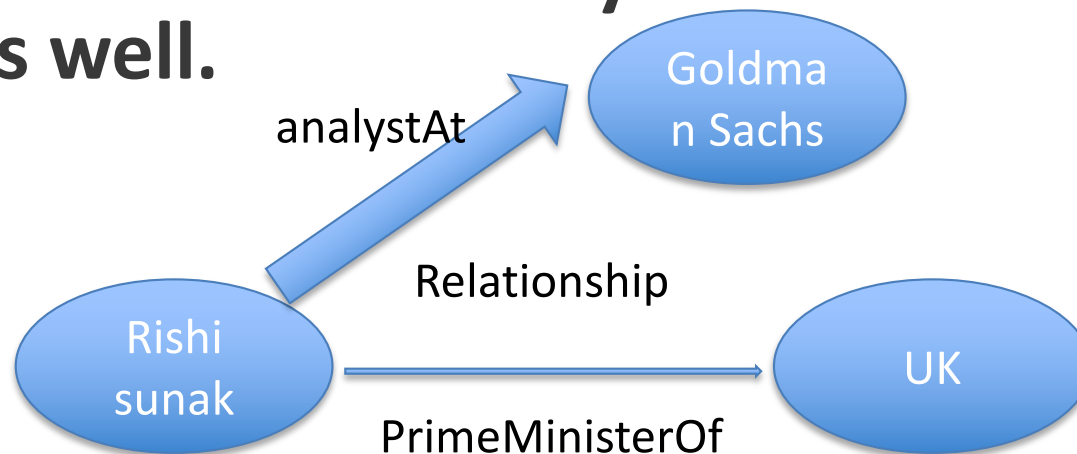
triple

- Node A and Node B here are two different entities.
- These nodes are connected by an edge that represents the relationship between the two nodes.
- A triple represents a couple of entities and a relation between them.



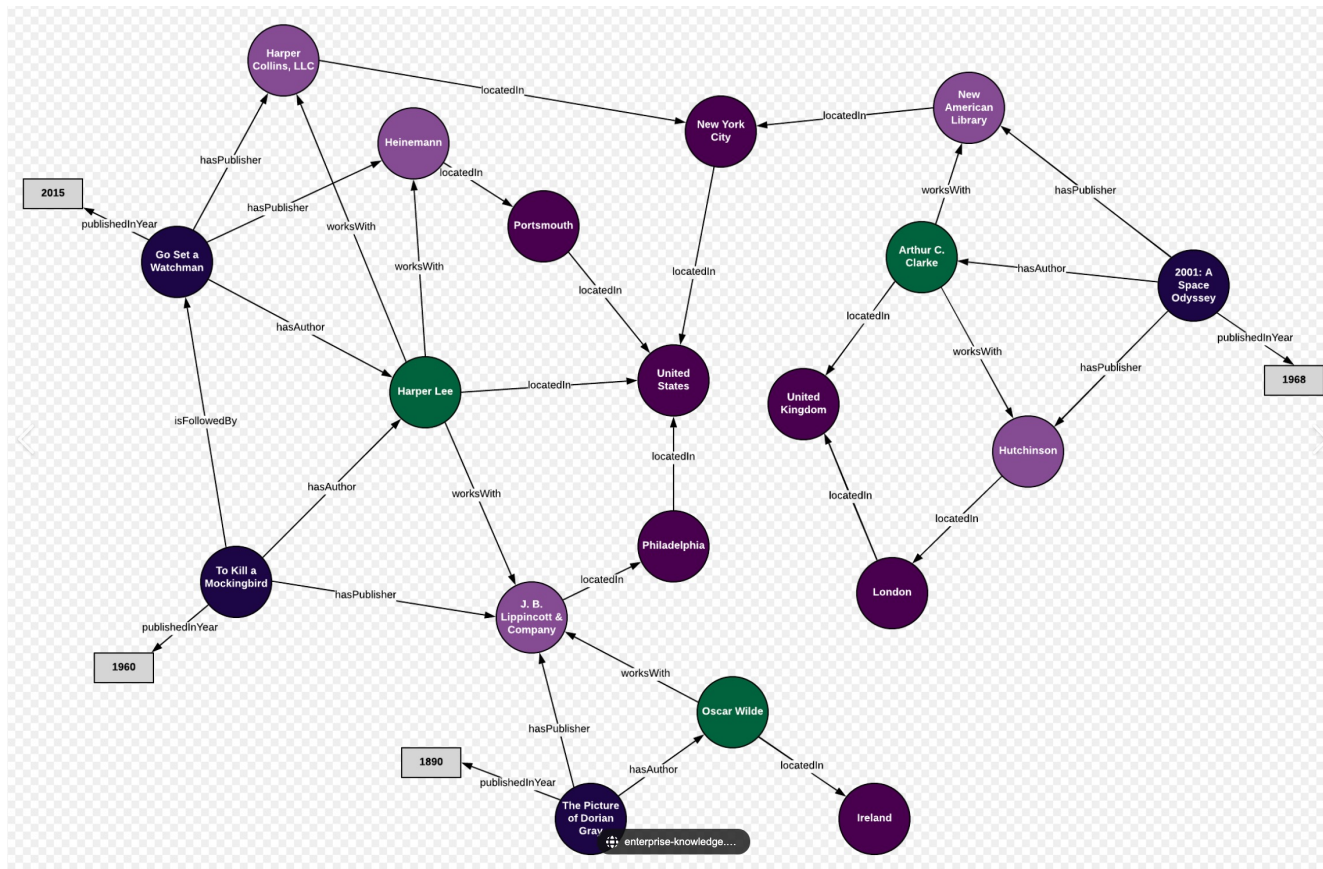
triple

- Node A = {Rishi Sunak} and Node B={UK} are two different entities.
- These nodes are connected by an edge that represents the relationship between the two nodes.
 - “Rishi Sunak” Prime Minister of “UK”
- **A node or an entity can have multiple relations as well.**



KG?

- Knowledge Graph's come in a variety of shapes and sizes.
- For example, the knowledge graph of Wikidata
- had 59,910,568 nodes by October 2019.
- Wikidata
 - Wikidata is a free and open knowledge base that can be read and edited by both humans and machines.
 - Wikidata acts as central storage for the **structured data** of its Wikimedia sister projects including Wikipedia, Wikivoyage, Wiktionary, Wikisource, and others.



Graph

- Relations form a graph
 - the nodes refer to the “real” data or contain some literal
 - how the graph is represented in machine is immaterial for now
- In a knowledge-graph information represented in a particular formal ontology
 - can be more easily accessible to automated information processing,
 - and how best to do this is an active area of research in computer science like data science/semantic web.

DBPedia

- **DBPedia** <http://dbpedia.org> —
 - Giving structure to **Wikipedia**
- The 2016-04 release of the DBpedia data set
 - 6M entities: *1.5M people, 810K places, 301K species, 275K organizations, 135K music albums, 106K films, 20K video games, ..*
 - consists of 9.5 billion RDF triples,
 - 1.3 billion were extracted from the English edition of Wikipedia and 5.0 billion from other language editions

DBPedia Entry Innsbruck

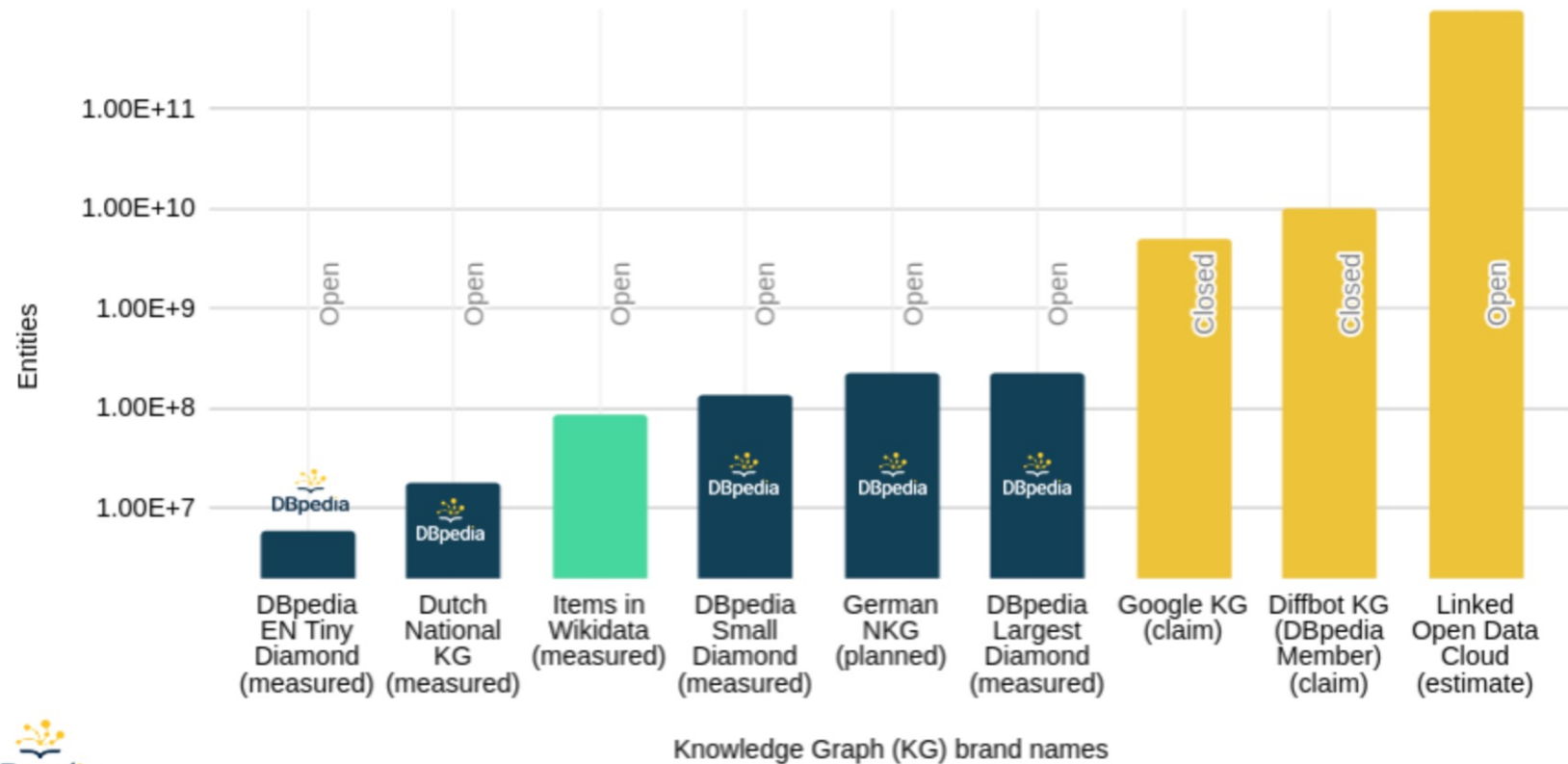
```
{{Infobox Town AT |
name = Innsbruck |
image_coa = InnsbruckWappen.png |
image_map = Karte-tirol-I.png |
state = [[Tyrol]] |
regbzkg = [[Statutory city]] |
population = 117,342 |
population_as_of = 2006 |
pop_dens = 1,119 |
area = 104.91 |
elevation = 574 |
lat_deg = 47 |
lat_min = 16 |
lat_hem = N |
lon_deg = 11 |
lon_min = 23 |
lon_hem = E |
postal_code = 6010-6080 |
area_code = 0512 |
licence = I |
mayor = Hilde Zach |
website = [http://innsbruck.at] |
}}
```

Innsbruck

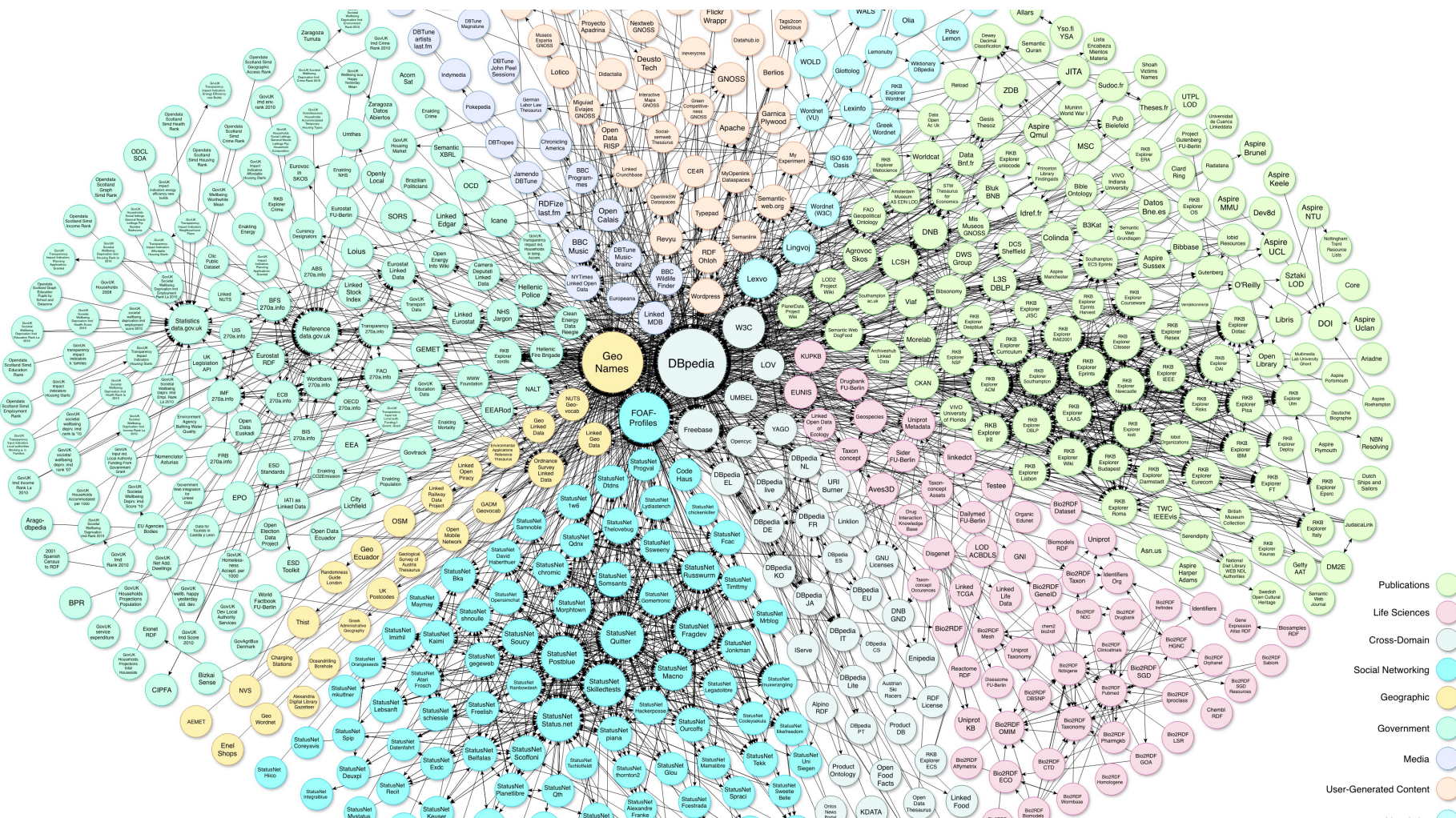



Country	Austria
State	Tyrol
Administrative region	Statutory city
Population	117,342 (2006)
Area	104.91 km²
Population density	1,119 /km²
Elevation	574 m
Coordinates	47°16' N 11°23' E 
Postal code	6010-6080
Area code	0512
Licence plate code	I
Mayor	Hilde Zach
Website	www.innsbruck.at 

Sep 2020: Current entity size of selected KGs (open and closed), log-scale

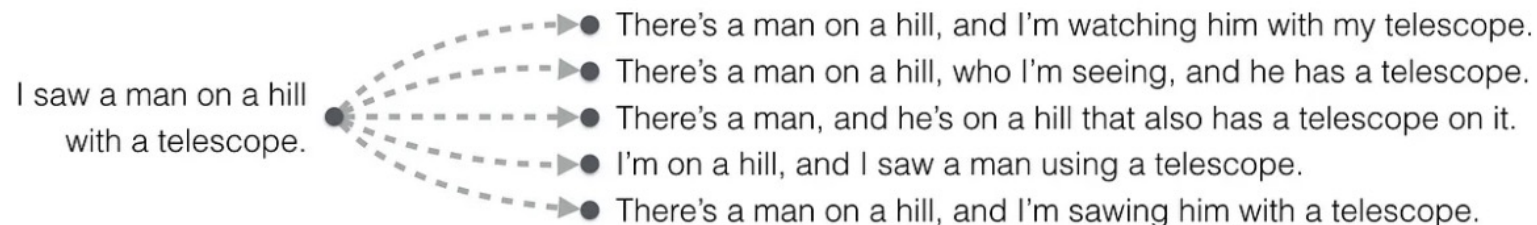


Linked Open Data



How to build KGs?

- Manually building KG is not scalable
- machines do not understand natural language.



Difficulties of language you say?

DATA TO KNOWLEDGE

Data to Knowledge

- Large amounts of text on the web
 - News reports, social media posts, wikipedia articles etc.
 - Scale and diversity of data
- How to extract knowledge from such data?

entity

- An entity is an object that exists.
- Named entity
 - a named entity is a real-world object, such as
 - a person, location, organization, product, etc.,
 - that can be denoted with a proper name.
- Named entity recognition (NER) is a
 - natural language processing technique that identifies
 - **predefined categories of objects in a body of text.**

Information extraction (IE) systems

- Find and understand limited relevant parts of texts
- Gather information from many pieces of text
- Produce a structured representation of relevant information:
 - relations (in the database sense),
 - a knowledge base
 - Goals:
 - Organize information so that it is useful to people
 - Put information in a semantically precise form that allows further inferences to be made by computer algorithms

Named-entity recognition (NER)

- is a sub-task of information extraction that
 - seeks to locate and classify ***named entities*** in text into pre-defined categories
- such as the names of
 - persons, organizations, locations, expressions of times, quantities, monetary values, percentages, etc.
- NER also known as
 - entity identification,
 - entity chunking or
 - entity extraction

What Is Entity Recognition (ER) ?

- **Identify** token of entity mentions in text, and **classify** them into predefined set of types of interest
 - Typing

Why **[Kobe]** is better than **[Jordan]**?

How long is a flight from **[Los Angeles]** to **[London]**?

What Is Entity Recognition and Typing (ER)

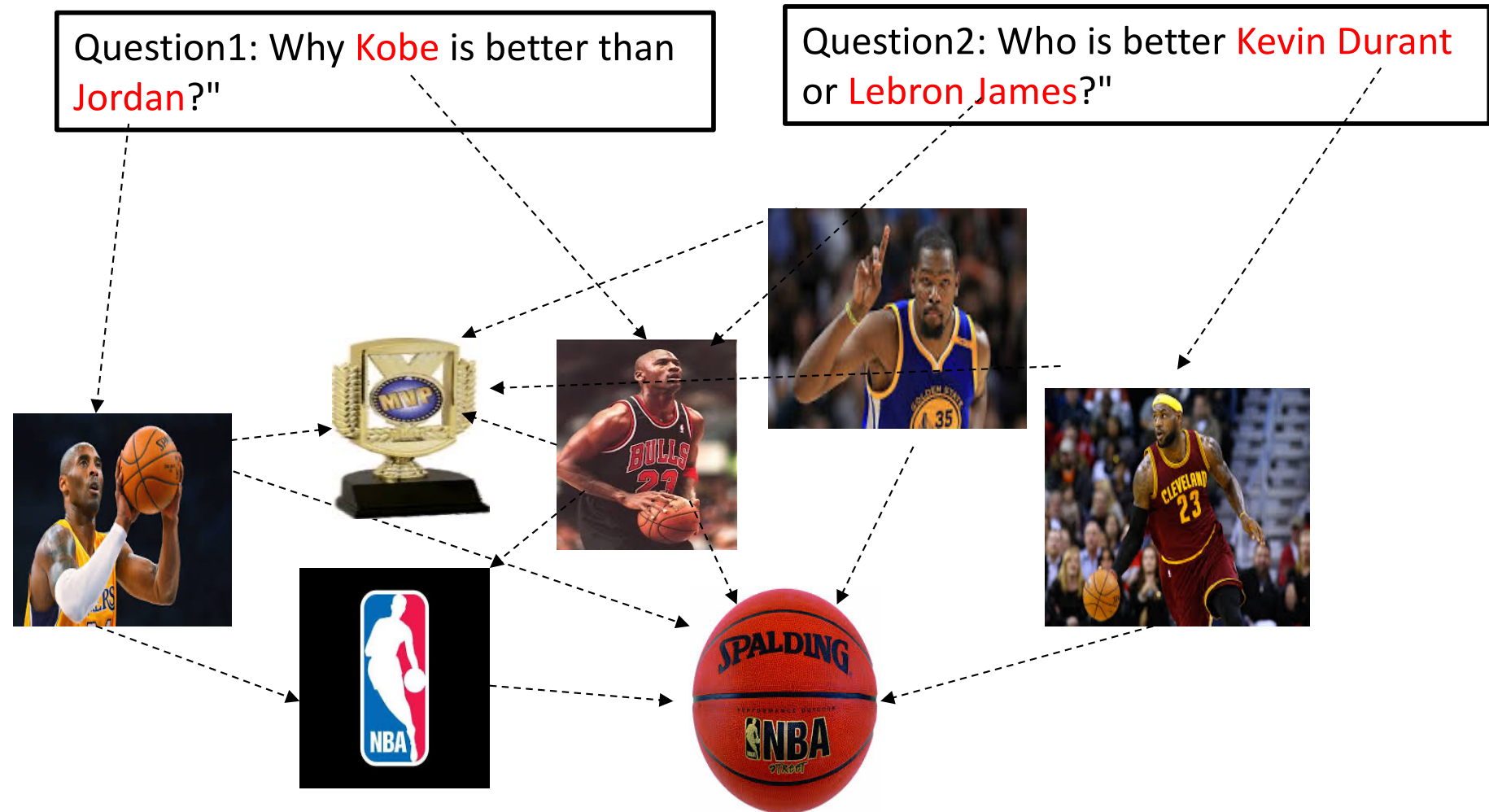
- **Identify** token spans of entity mentions in text, and **classify** them into predefined set of types of interest

Why [Kobe] is better than [Jordan]?

How long is a flight from [Los Angeles] to [London]?

PERSON
LOCATI
ON

Example: Linking Entities to Knowledge Base

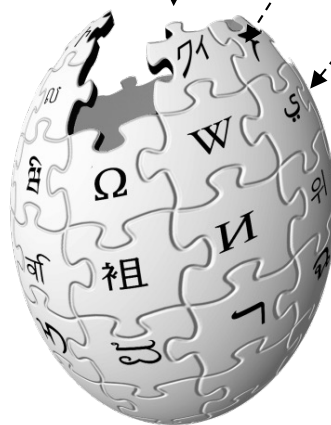


“Entities” are what a large part of our knowledge is about

Example: Linking Entities to Knowledge Base

Question1: Why **Kobe** is better than **Jordan**?"

Question2: Who is better **Kelvin Durant** or **Lebron James**?"



Link entity mentions to knowledge base entries for in-depth entity information

WIKIPEDIA

The Free Encyclopedia

https://dbpedia.org/page/Kobe_Bryant

<https://www.wikidata.org/wiki/Q25369>

Approaches

- NER systems have been created that use
 - linguistic grammar-based techniques as well as
 - Hand-crafted grammar-based systems typically obtain better precision,
 - but at the cost of lower recall and months of work by experienced computational linguists .
 - statistical models such as machine learning.
 - Statistical NER systems typically require a large amount of manually annotated training data.
 - Semi-supervised approaches have been suggested to avoid part of the annotation effort.

NB classifier

- Use conventional classification algorithms to classify substrings of document as “to be extracted” or not.
- In some simple but compelling domains, this naive technique is remarkably effective.

NER Tools

- spaCy NER Model :
 - A free and an open-source library, spaCy
 - advanced Natural Language Processing (NLP) much simpler in Python.
- Stanford Named Entity Recognizer :
 - Stanford NER is a Named Entity Recognizer, implemented in Java.
 - It provides a default trained model for recognizing chiefly entities like Organization, Person and Location.
- Stanford NER is also referred to as a
 - CRF (Conditional Random Field) Classifier as
 - Linear chain Conditional Random Field (CRF) sequence models have been implemented in the software.
 - We can train our own custom models with our own labeled dataset for various applications.

Why NER ?

Application Scenarios

- How is it used?
 - HR Department across companies is to evaluate a gigantic pile of resumes to shortlist candidates.
 - To add to their burden, resumes of applicants are often excessively populated in detail, of which, most of the information is irrelevant to what the evaluator is seeking.
 - NER model could facilitate evaluation of resumes at a quick glance, thereby simplifying the effort required in shortlisting candidates among a pile of resumes.
- A search engine algorithm,
 - instead of searching for an entered query across the millions of articles and websites online,
 - a more efficient approach would be to run an NER model on the articles once and
 - store the entities associated with them permanently.
 - The key tags in the search query can then be compared with the tags associated with the website articles for a quick and efficient search.

Knowledge Graphs

- The idea behind **Knowledge Graphs (KG)** to answer questions like
 - *Machine Learning and Data Science are related or not?*
 - *Noam Chomsky is an American Linguist?*
 - *What about the height of Mount Makalu?*

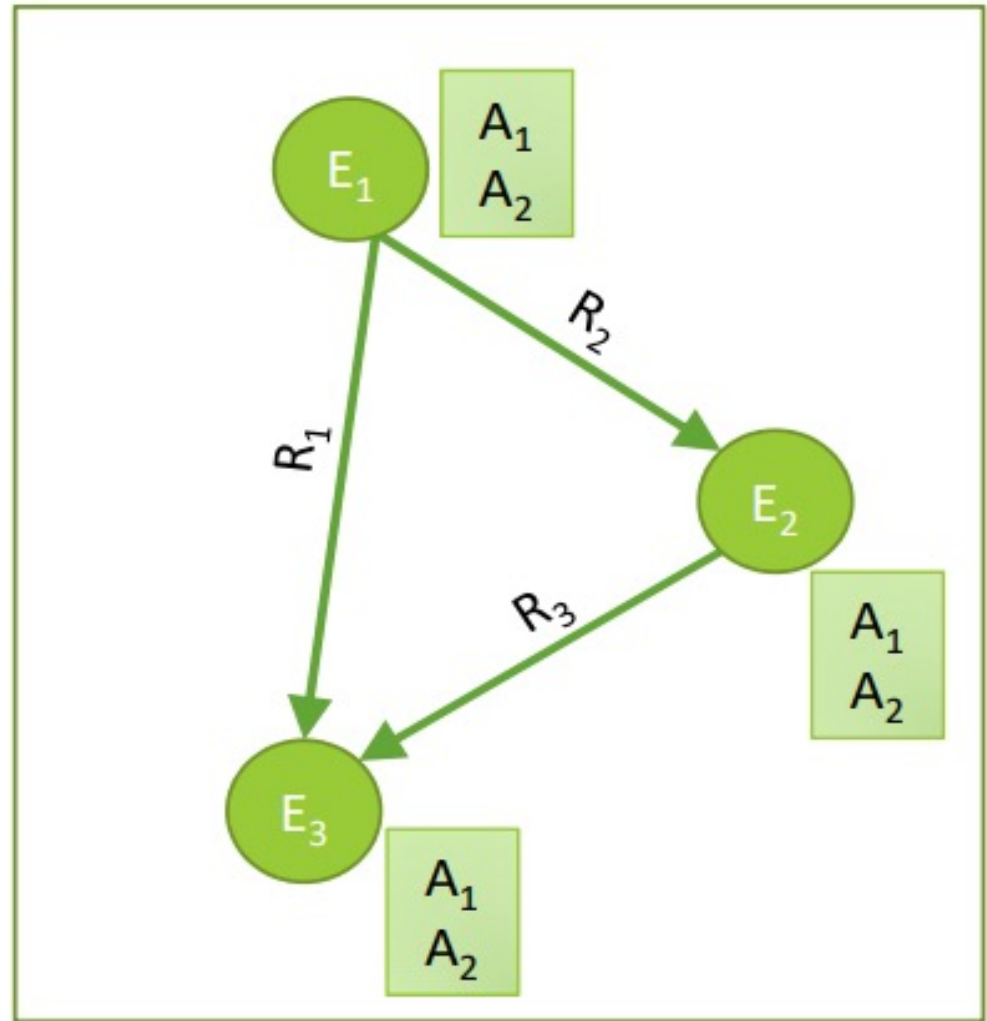
PART II

First look at an example (hack); then look at research methods (more scalable/robust methods)

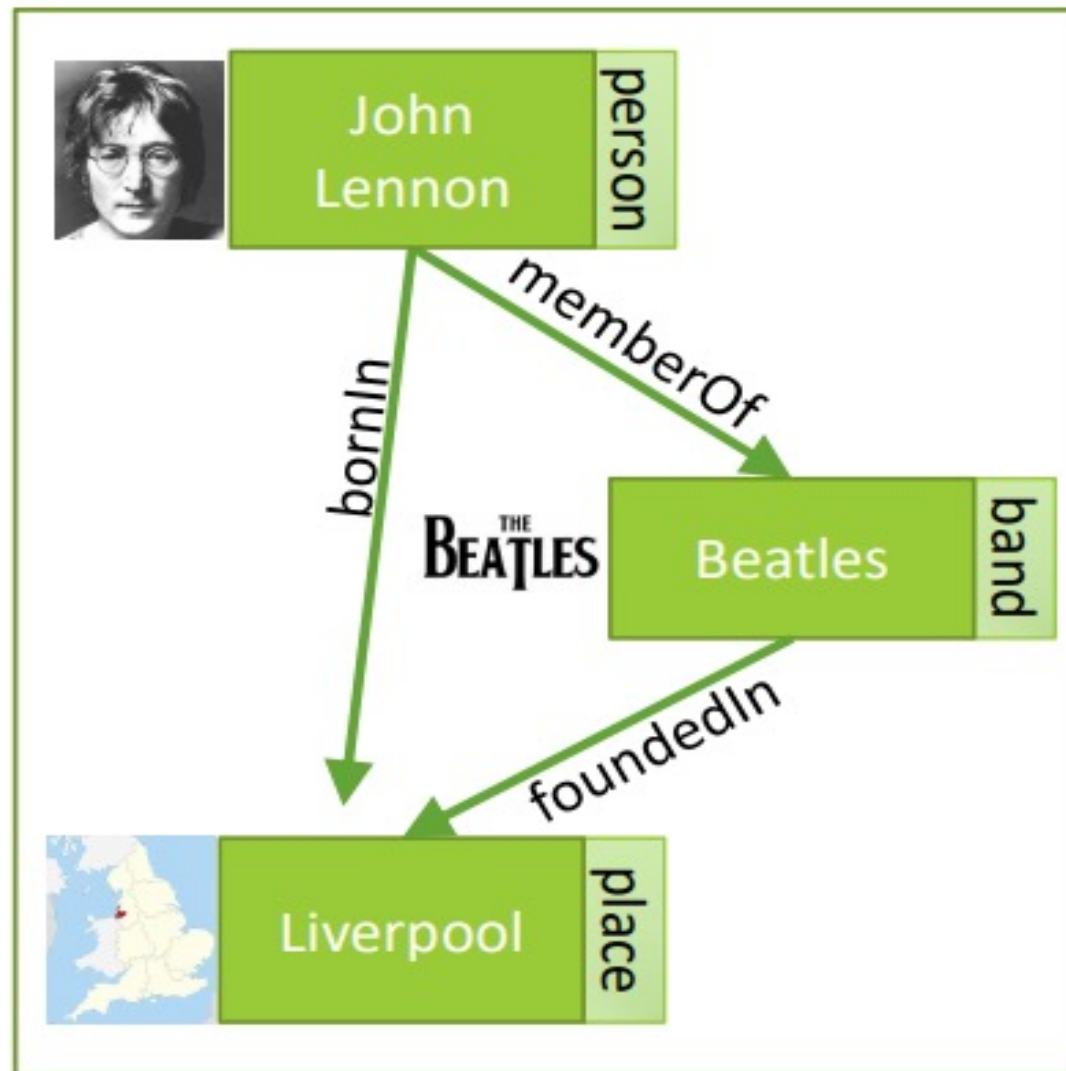
KNOWLEDGE GRAPH

Knowledge Graph

- Captures entities, attributes, and relationships
- Nodes are entities
- Nodes are labeled with attributes (e.g., types)
- Typed edges between two nodes capture a relationship between entities



Entities & relationships

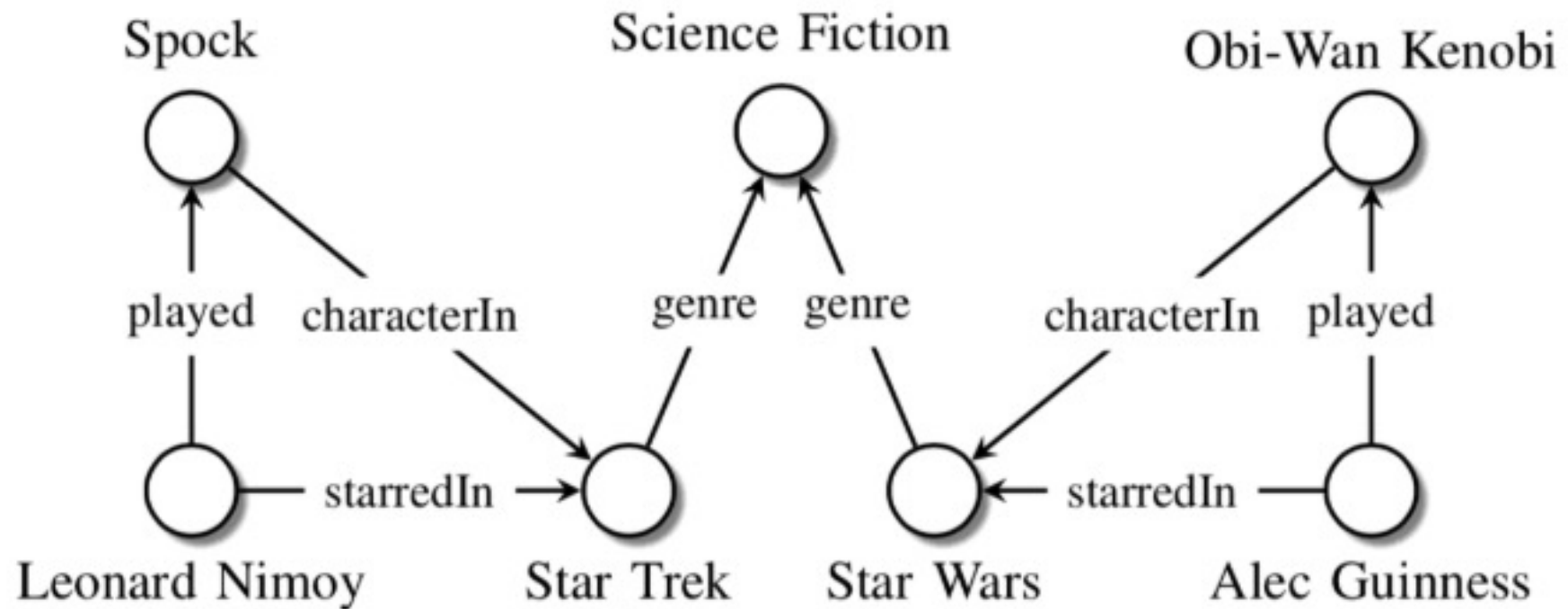


Example

- *Leonard Nimoy was an actor who played the character Spock in the science-fiction movie Star Trek*
- The triples (facts) that can be extracted from this sentence are :

Subject	predicate	object
LeonardNimoy	profession	Actor
LeonardNimoy	starredIn	StarTrek
LeonardNimoy	played	Spock
Spock	CharacterIn	StratTrek
StarTrek	genre	ScienceFiction

As a knowledge Graph!



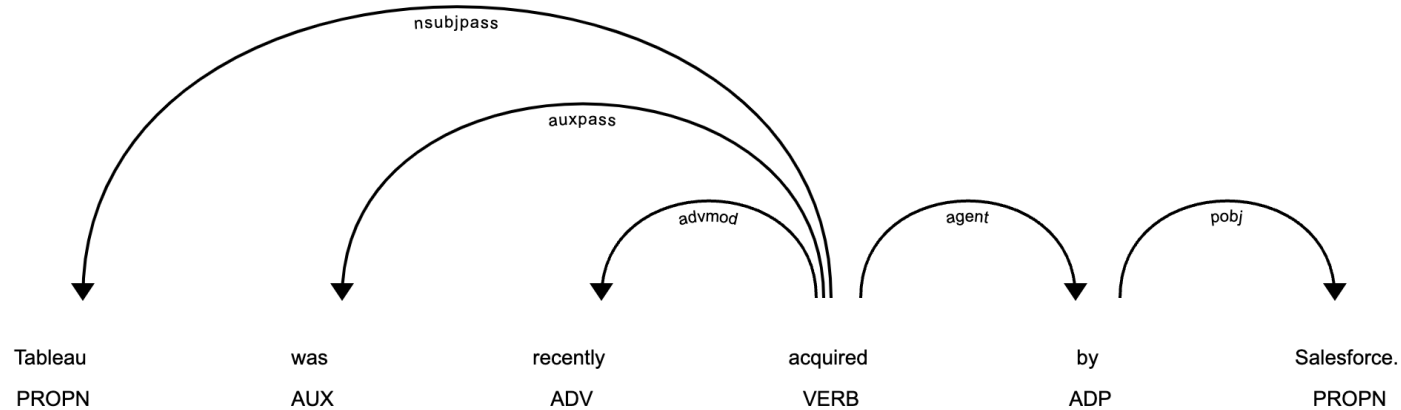
<https://arxiv.org/1503.00759.pdf>

Maximillian Nickel, Kevin Murphy, Volker Tresp, Evgeny Gabrilovich; Google

Dependency Parsing

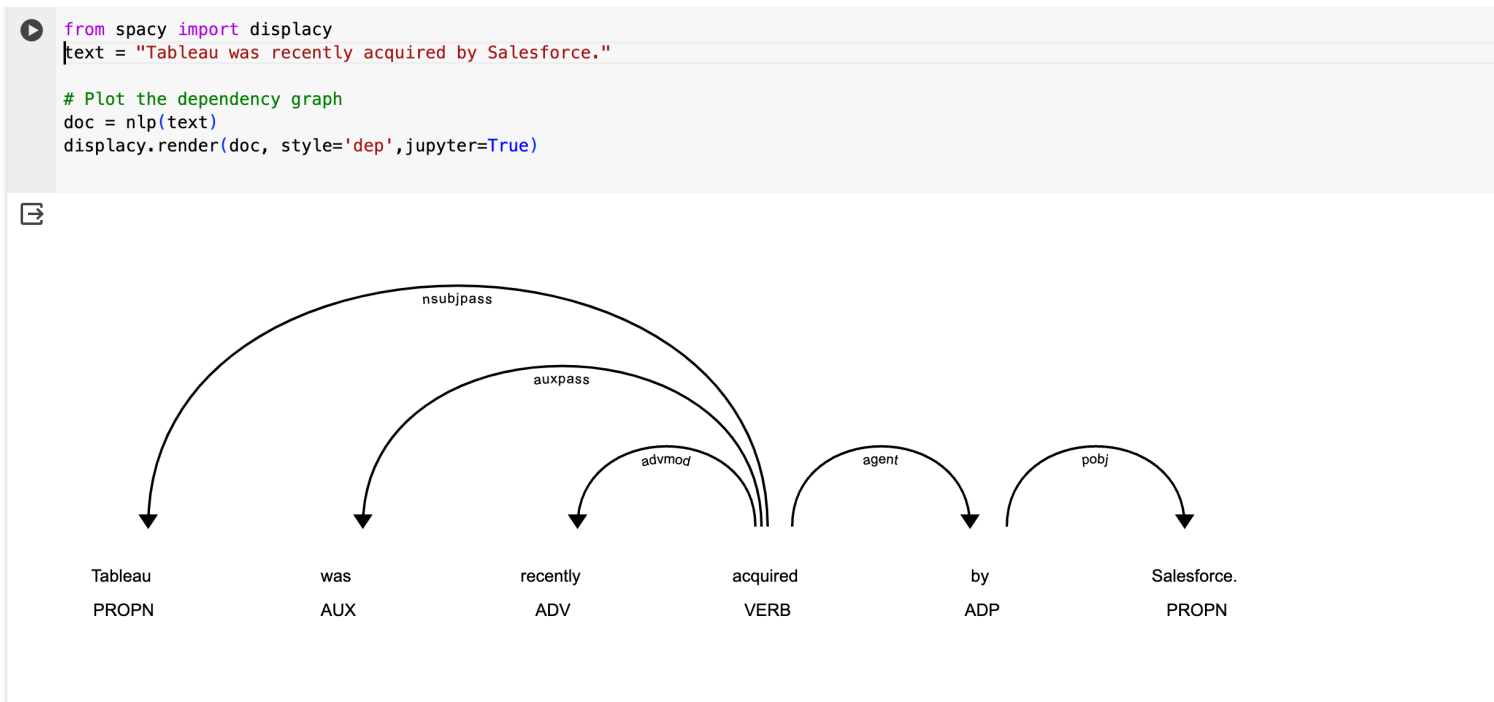
```
from spacy import displacy
text = "Tableau was recently acquired by Salesforce."

# Plot the dependency graph
doc = nlp(text)
displacy.render(doc, style='dep', jupyter=True)
```



What did we do?

- **Extraction of SPO triples (facts) from text.**
 - Uses **Natural Language Processing (NLP)** techniques like **dependency parsing**.



What did we do?

- **Extraction of SPO triples** (facts) from text.
- **Entity Recognition & Linking:**
 - This is the step that maps Leonard N, L Nimoy, Leo Nimoy, etc. all to a single entity i.e. LeonardNemoy.
- **DBpedia**, the structured data store of Wikipedia is used a single global storage of all the entities,.
 - So Leonard Nemoy would get mapped to
 - [http://dbpedia.org/page/Leonard Nimoy](http://dbpedia.org/page/Leonard_Nimoy).

Subject	predicate	object
LeonardNimoy	profession	Actor
LeonardNimoy	starredIn	StarTrek
LeonardNimoy	played	Spock

How to make a knowledge graph?

- Let's look at a sample text below!
- *“Indian tennis player Sumit Nagal moved up six places from 135 to a career-best 129 in the latest men's singles ranking. The 22-year-old recently won the ATP Challenger tournament. He made his Grand Slam debut against Federer in the 2019 US Open. Nagal won the first set.”*

Sentence Segmentation

step 1

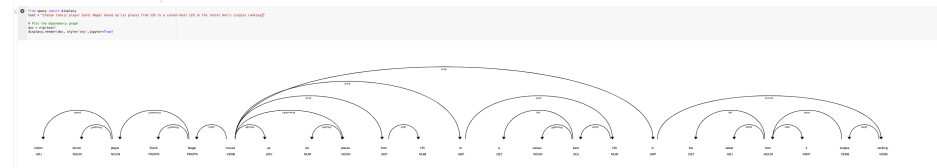
- The first step in building a knowledge graph is to
 - split the text document or article into sentences.
- Let's split the paragraph into sentences:
 - *Indian tennis player Sumit Nagal moved up six places from 135 to a career-best 129 in the latest men's singles ranking*
 - ***The 22-year-old recently won the ATP Challenger tournament***
 - *He made his Grand Slam debut against Federer in the 2019 US Open*
 - ***Nagal won the first set***

For simplicity

- *we will shortlist only those sentences in which there is exactly 1 subject and 1 object.*

```
nlp = spacy.load('en_core_web_sm')  
  
doc = nlp("Indian tennis player Sumit Nagal moved up six places from 135 to a career-best 129 in the latest men's singles ranking.")  
  
for tok in doc:  
    print(tok.text, "\t", "...", "\t", tok.dep_)
```

Indian	...	amod
tennis	...	compound
player	...	compound
Sumit	...	compound
Nagal	...	nsubj
moved	...	ROOT
up	...	advmod
six	...	nummod
places	...	npadvmod
from	...	prep
135	...	pobj
to	...	prep
a	...	det
career	...	npadvmod
-	...	punct
best	...	amod
129	...	pobj
in	...	prep
the	...	det
latest	...	amod
men	...	poss
's	...	case
singles	...	nsubj
ranking	...	pcomp
.	...	punct



Subject & object

step 2

- 2nd and 4th sentence
- In the second sentence, “22-year-old” is the subject, and the object is “ATP Challenger tournament”.
- In the fourth sentence, the subject is “Nagal” and “first set” is the object:

Sentence	Subject	Object
The 22-year-old recently won ATP Challenger tournament.	22-year-old	ATP Challenger tournament
Nagal won the first set.	Nagal	first set

Entities Extraction

step 3

- The extraction of a single word entity from a sentence is not a tough task!
- We can easily do this with the help of parts of speech (POS) tagging systems (e.g., Spacy).
 - The nouns and the proper nouns would be our entities.
- *The Rule*
 - ***extract the subject/object along with its modifiers***
 - ***and also extract the punctuation marks between them.***

SpaCy Output

```
▶ import spacy
nlp = spacy.load('en_core_web_sm')

doc = nlp("The 22-year-old recently won ATP Challenger tournament.")

for tok in doc:
    print(tok.text, "...", tok.dep_)
```

```
⇒ The ... det
22 ... nummod ←
- ... punct
year ... npadvmod ←
- ... punct
→ old ... nsubj
recently ... advmod
won ... ROOT
ATP ... compound
Challenger ... compound
tournament ... dobj
. ... punct
```

- * The subject (*nsubj*) in this sentence is “old”.
- * Our aim is to extract “22-year-old”.
 - dependency tag of “22” is *nummod* and
 - “year” is *npadvmod*
- means modifier of “old”.

we need to define a rule to extract such entities.
22-year-old

SpaCy Output

```
▶ import spacy
nlp = spacy.load('en_core_web_sm')

doc = nlp("The 22-year-old recently won ATP Challenger tournament.")

for tok in doc:
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```

⇒ The ... det
22 ... nummod ←
- ... punct
year ... npadvmod ←
- ... punct
old ... nsubj ←
recently ... advmod
won ... ROOT
ATP ... compound
Challenger ... compound
tournament ... dobj
. ... punct

- * The subject (*nsubj*) in this sentence is “old”.
- * Our aim is to extract “22-year-old”.
 - dependency tag of “22” is *nummod* and
 - “year” is *npadvmod*
- means modifier of “old”.

we need to define a rule to extract such entities.
22-year-old

Compound words

step 4

- Compound words are those words that collectively form a new term
 - with a different meaning.
- look at the object (*dobj*) in the sentence.
 - It is just “tournament” instead of
 - “ATP Challenger tournament”.
 - These are compound words
- Update rule to
 - **extract the subject/object along with its modifiers, compound words**
 - **and also extract the punctuation marks between them.**

Challenges in KE

- Documents contain inaccurate, outdated, incomplete or hypothetical information or
 - creative languages used in documents often difficult to interpret
- In reality, ... candidates facts extracted
 - Often miss key facts
 - Include spurious outputs
 - Compromising the usefulness of these extractions

Extract Relations

step 5

- We need edges to connect the nodes (entities) to one another
 - These edges are the ***relations*** between a pair of nodes.
- Let's go back to the example
 - We shortlisted 2(2nd and 4th) sentences to build a knowledge graph

2nd - The 22-year-old recently won the ATP Challenger tournament

4th - Nagal won the first set

“won” verb (ROOT)

```
import spacy
nlp = spacy.load('en_core_web_sm')

doc = nlp("The 22-year-old recently won ATP Challenger tournament.")

for tok in doc:
    print(tok.text, "\t", "...", "\t", tok.dep_)
```



The	...	det
22	...	nummod
-	...	punct
year	...	npadvmod
-	...	punct
old	...	nsubj
recently	...	advmod
won	...	ROOT
ATP	...	compound
Challenger	...	compound
tournament	...	dobj
.	...	punct

```
doc = nlp("Nagal won the first set.")

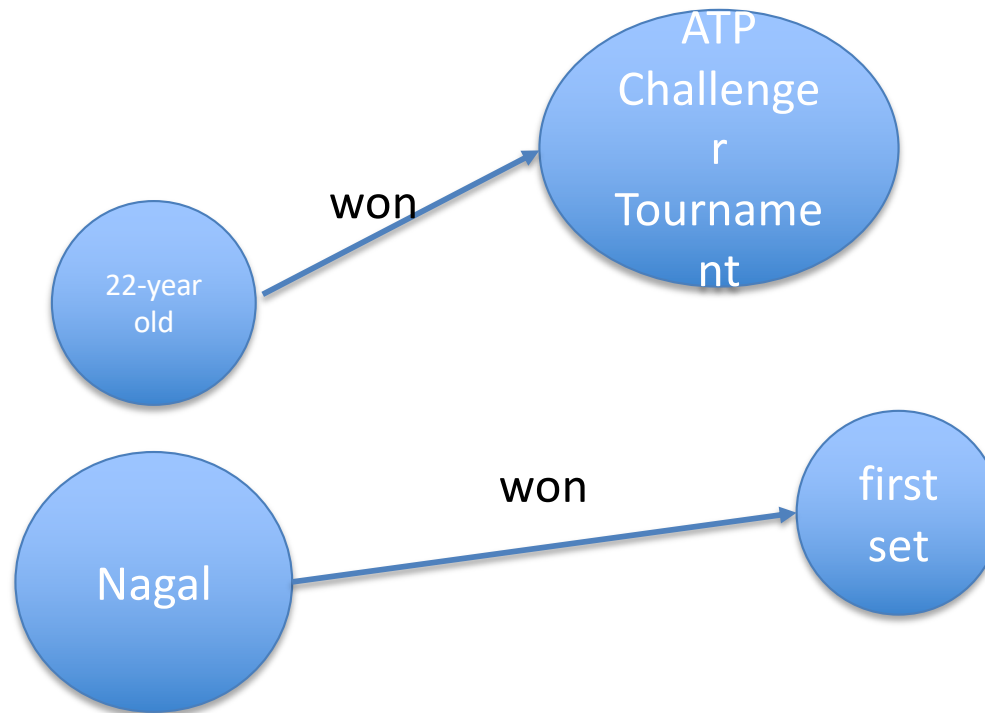
for tok in doc:
    print(tok.text, "...", tok.dep_)
```



Nagal	...	nsubj
won	...	ROOT
the	...	det
first	...	amod
set	...	dobj
.	...	punct

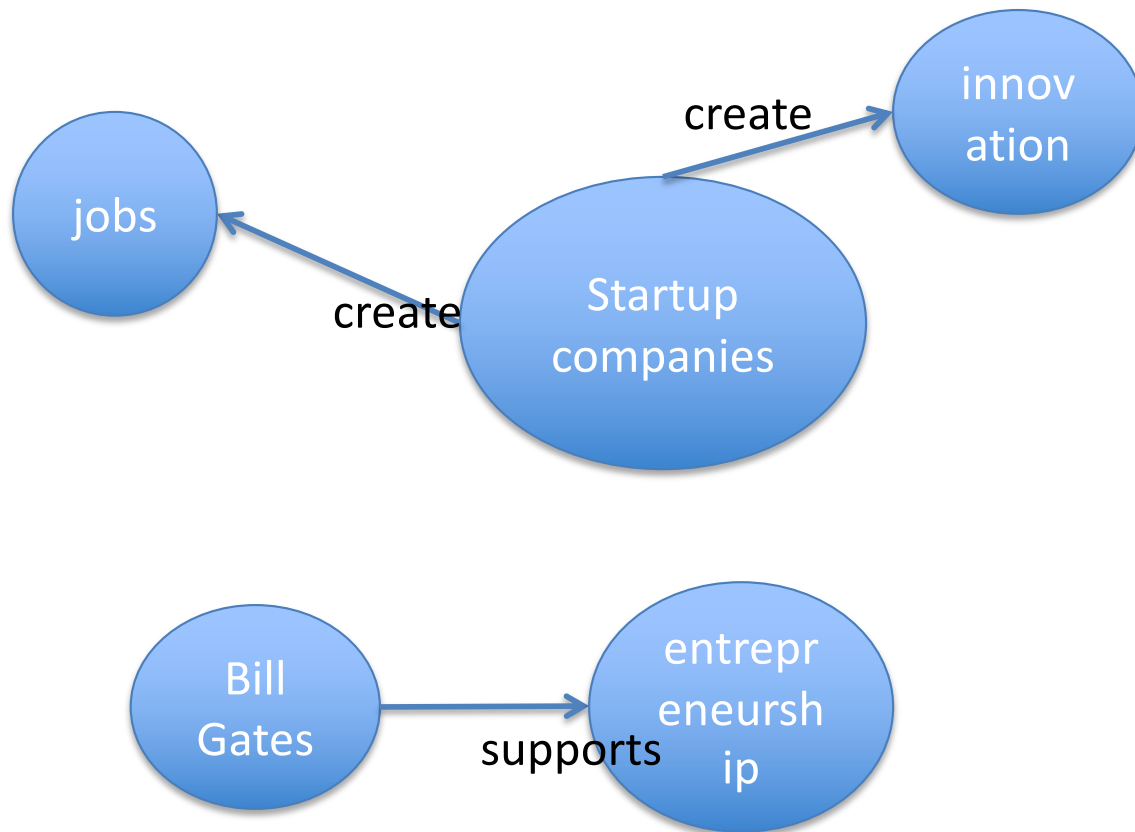
triplets

- **triples: subject, object, and and predicate**
 - (or *entity-attribute-value*).



Another example

- *Startup companies create jobs and innovation.
Bill Gates supports entrepreneurship.*



Graph Construction

step 6

- the fact represented as an SPO conveys that
 - the Subject is related to the Object
 - through the relationship described by the Predicate.
- storing the SPO triples on a Graph Database
 - Remove any ambiguities
- Process the Graph to achieve things like
 - *filling missing links, clustering entities, etc.*

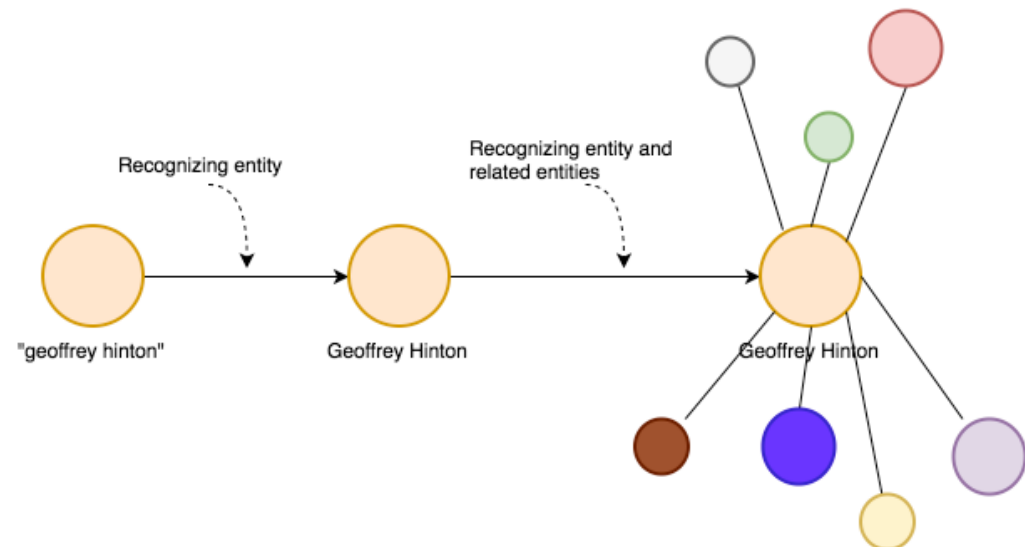
Resource Description Framework (RDF)

- RDF is a standard model for **data interchange** on the Web.
- RDF has features that facilitate data merging even if the underlying schemas differ, and
 - it specifically supports the evolution of schemas over time without requiring all the data consumers to be changed.
- RDF is one of the three foundational Semantic Web technologies,
 - RDF is the data model of the Semantic Web.

```
1 <?xml version="1.0"?>
2 <rdf:RDF
3     xmlns:rdf="http://www.w3.org/1999/02/22-rdf-syntax-ns#"
4     xmlns:si="http://www.glasgow.ac.uk/singapore/">
5     <rdf:Description rdf:about="http://www.glasgow.ac.uk">
6         <si:title>University of Glasgow Singapore</si:title>
7     </rdf:Description>
8 </rdf:RDF>
```

example

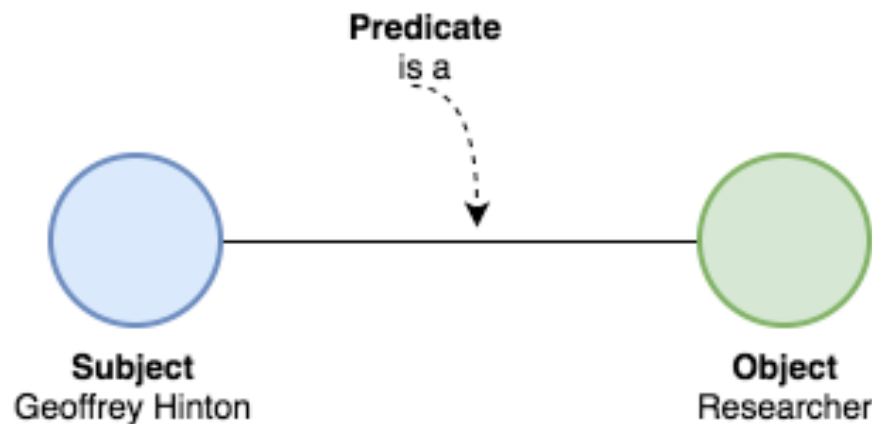
- Let us say we have string “Geoffrey hinton”.
- first the system recognizes the that string actually meant the person Geoffrey Hinton.
 - And then it will recognize the related entities to that person.



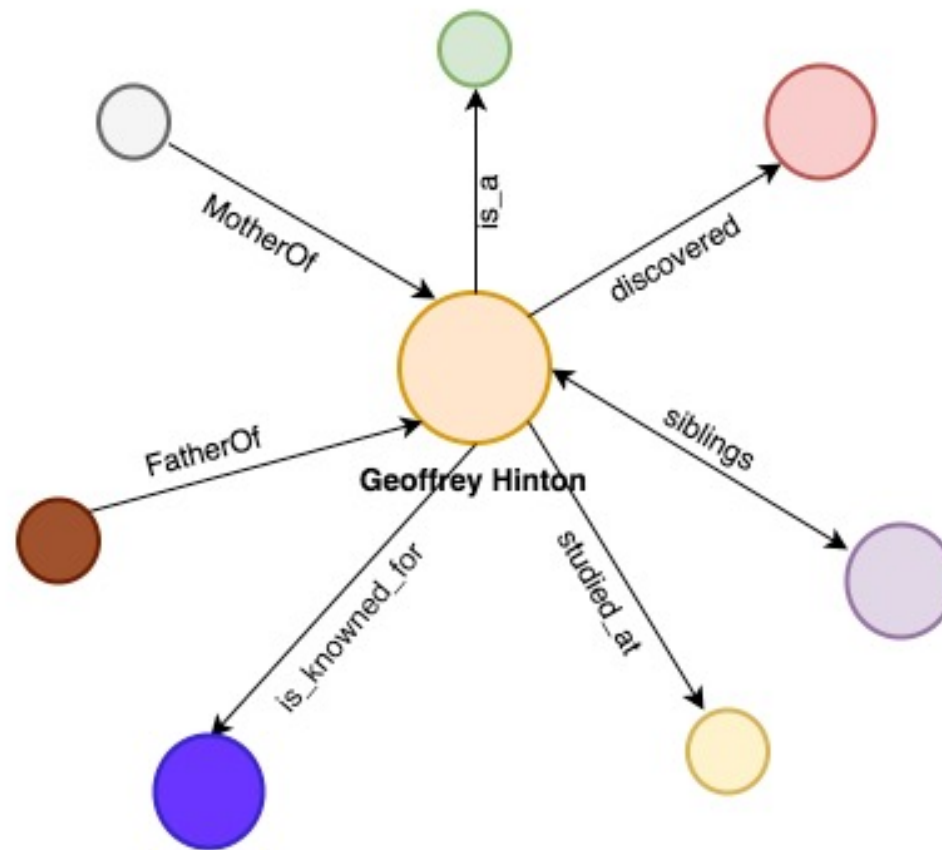
Then we have some entities that are related to Geoffrey but we don't know what they are yet.

knowledge-graph & the triples

- **subject** <Geoffrey Hinton> related to
- the **object** <Researcher> by
- the **predicate** <is a>.



To have a triple we need a subject and object, and a predicate linking the two.



Now we have named relationships where we know what type of connection we have for our main entity.

Knowledge Graph

- Contain facts in the knowledge base
 - Entities are represented as nodes
 - Attributes of each entity are node labels
 - Relationship between two or more entities represented as edges
- To build knowledge graph
 - Input entities, labels, and relationships produced by an information extraction system
 - In real-life
 - Noise, ambiguity, errors limits the usefulness of extraction graph

Case study

scalable & robust

- Academia
 - Using Semantics & Statistics to Turn Data into Knowledge,
 - Pujara et al (2014),
 - In AAAI
- Google approach (another)
 - Xin Luna Dong, Evgeniy Gabrilovich, Jeremy Heitz, Wilko Horn, Ni Lao, Kevin Murphy, Thomas Strohmann, Shaohua Sun, Wei Zhang
 - Knowledge Vault: A Web-Scale Approach to Probabilistic Knowledge Fusion;
 - KDD 2014

Extraction Process

- Vast variety of features
- Statistics
 - Ranging from the characteristics of web pages
 - Reliability of certain techniques used
- Use the above & modest amount of training data
 - Employ heuristics or learned prediction functions to assign candidate scores to each candidate fact

Using Semantics & Statistics to Turn Data into Knowledge,
Pujara et al (2014),

Knowledge Graph Identification

- Process of
 - Entity resolution
 - Collective classification
 - Link prediction
- Jointly inferring the entities, labels and relationships in a graph
- Incorporate ontological constraints between facts during inference

Describe the role of Entity resolution, Collective classification, Link prediction in knowledge graph identification

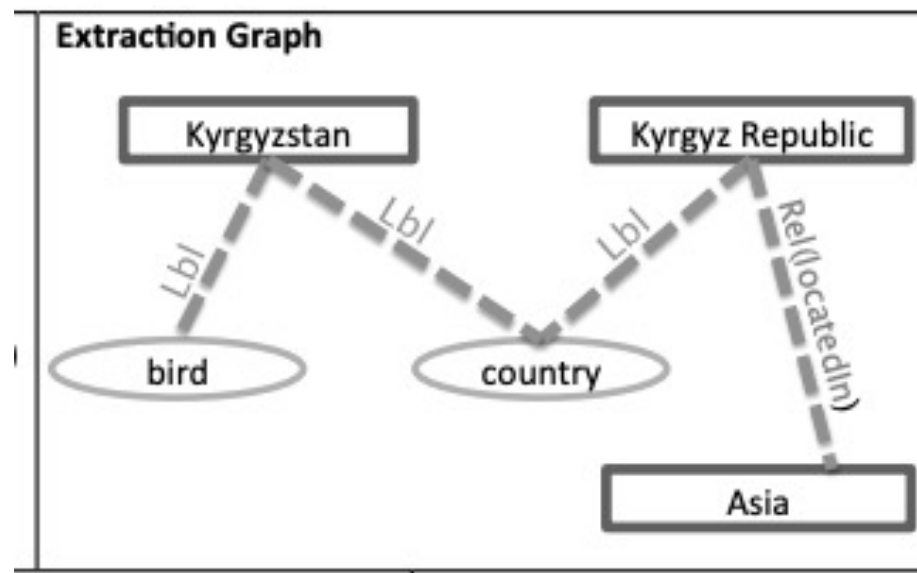


- Features derived from text to measure the quality of candidate facts
- Many semantic dependencies between facts
- Let us look at Kyrgistan
 - **Country or region: Kyrgyzstan**
 - Other names: Kirghizia, Kyrgyz Republic

Statistics & Extractions from NELL

- [.75, .5] bird(Kyrgistan)
- [.60, .80] country(Kyrgystan)
- [.90,.55] country(Kyrgyz Republic)
- [.80, .65] locatedIn(Kyrgyz Republic, Asia)

NELL - Never ending
language learning tools



Entity resolution (ER)

- Many textual references that initially look different may refer to the same real-world entity
- Example (variants or misspellings) in NELL
 - Kyrghystan
 - Kyrgyzstan
 - Kyrgysatn
 - Kyrgyz republic
 - Kyrgyzstan
 - Kyrgistan
- Extraction graph
 - incorrectly correspond to different nodes
- ER
 - Determine coreferent entities in the knowledge graph
 - Producing a consistent set of labels and relations for each resolved node

what is entity resolution? how can we do it?

Quality of KG

- Co-referent entities
 - Kyrgyzstan and Kyrgyz Republic as the same entity
 - Should have same labels and relationships
 - Will improve the quality of knowledge graph
- Multiple ontological constraints
 - Bird and country are mutually exclusive
 - Relation locatedIn
 - Mapping from a domain of countries to a range of continents

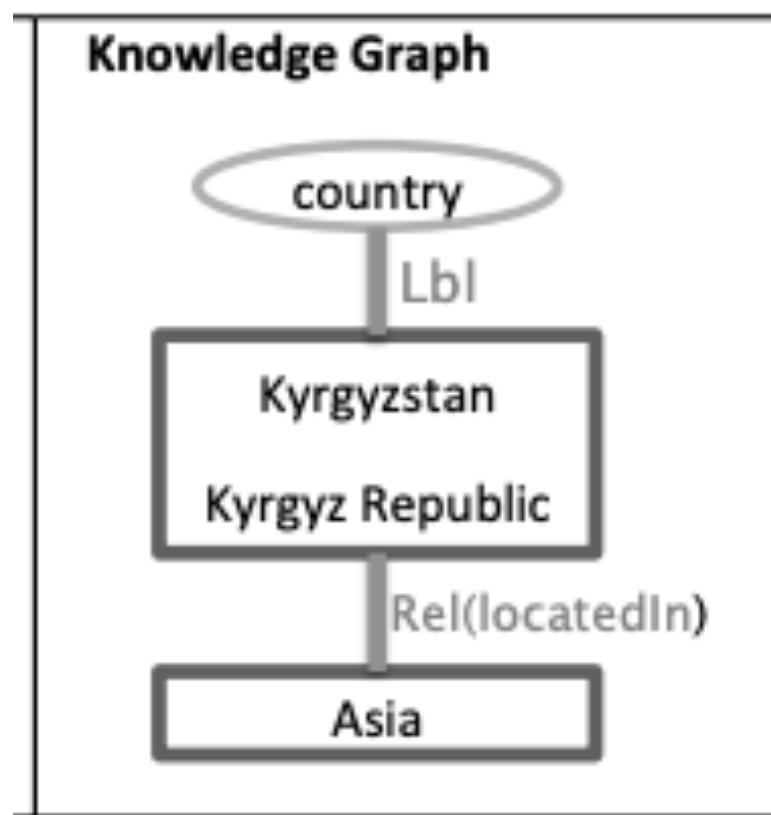
Collective Classification

- Kyrgystan as a bird in NELL
- Ontological constraints
 - Unlikely both bird and country
- Bishkent capital of Kyrgystan
 - Country
- Using the labels of related entities in the knowledge graph
 - Correct label of an entity
- Collective Classification
 - Label nodes in
 - Considering ontology and neighboring labels

Link prediction

- Many different facts relating to location
- Example
 - Kyrgyzstan is located in Kazakhstan
 - Kyrgyzstan is located in Russia
 - Kyrgyzstan is located in former Soviet Union
 - Kyrgyzstan is located in Asia
 - Kyrgyzstan is located in US
- Some are true & others are contradictory or false
- Predict links
 - Ontological information and the rest of the inferred structure

Knowledge Graph



Summary Poojara work

- Bootstrap using NELL
- Entity Resolution
- Collective Classification
- Link Prediction